

# DATA ANALYSIS INTERNSHIP

## Day 21 Report

### Power BI Relationships and Conditional Columns

#### Objective

The objective of this session was to learn how to create relationships between tables in Microsoft Power BI and how to use Conditional Columns to generate new columns based on conditions.

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#### Creating Relationships in Power BI

During this session, two student datasets were imported into Power BI:

#### Tables Used

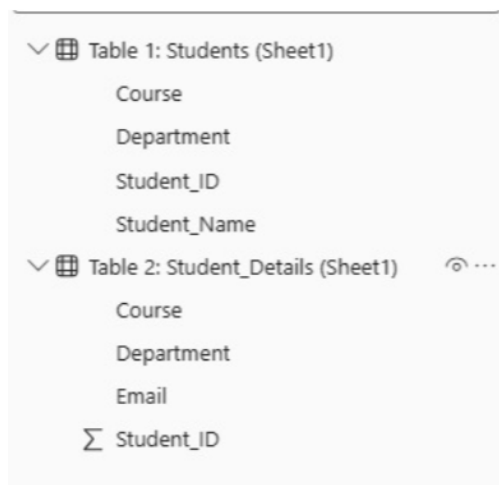
1. Students Table
2. Student\_Details Table

The relationship between the tables was created using common columns such as:

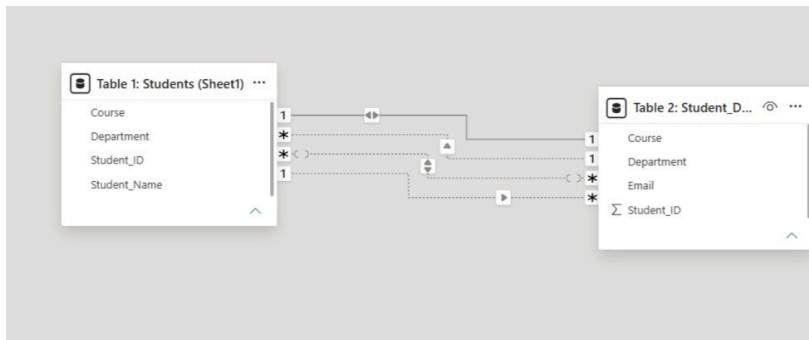
Student\_ID

Department

Course



#### Types of Relationships Created



## 1. One-to-One Relationship

### Connection:

Student\_ID ↔ Student\_ID

### Explanation:

One student record matches one detail record.

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## 2. One-to-Many Relationship

### Connection:

Department → Course

### Explanation:

One department contains multiple courses.

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## 3. Many-to-One Relationship

### Connection:

Course → Department

### Explanation:

Many courses belong to one department.

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## 4. Many-to-Many Relationship

### Connection:

Department ↔ Course

### Explanation:

Multiple departments can contain multiple courses.

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**Conditional Column**

A Conditional Column was created to convert short department names into full department names.

**Steps Performed**

- 1. Open:

Transform Data

- 2. Go to:

Add Column → Conditional Column

- 3. Create conditions based on Department values.

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

FullFormDepartment

Column Name

Operator

Value

Output

If

Department

equals

ABC123CSE

Then

ABC123ComputerScienceEngineering

Else If

Department

equals

ABC123IT

Then

ABC123Information Technology

\*\*\*

Add Clause

Else

ABC123

nd Communication Engineering

OK

Cancel

**Conditional Column Example**

**Department FullFormDepartment**

|     |                                           |
|-----|-------------------------------------------|
| CSE | Computer Science Engineering              |
| IT  | Information Technology                    |
| ECE | Electronics and Communication Engineering |

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**Conditions Used**

If:

Department = CSE

Then:

ComputerScienceEngineering

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**Else If:**

Department = IT

Then:

Information Technology

---

**Else:**

Electronics and Communication Engineering

This created a new column named:

FullFormDepartment

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## Output

The new conditional column successfully displayed full department names automatically based on department short forms.

| 123 Student_ID       | A <sup>B</sup> <sub>C</sub> Student_Name | A <sup>B</sup> <sub>C</sub> Department | A <sup>B</sup> <sub>C</sub> Course | ABC 123 FullFormDepartment            |
|----------------------|------------------------------------------|----------------------------------------|------------------------------------|---------------------------------------|
| 7 distinct, 7 unique | 7 distinct, 7 unique                     | 1 distinct, 0 unique                   | 3 distinct, 1 unique               |                                       |
| 101                  | Abi                                      | CSE                                    | Python                             | ComputerScience Engineering           |
| 102                  | Arun                                     | CSE                                    | Java                               | ComputerScience Engineering           |
| 103                  | Priya                                    | IT                                     | DBMS                               | Information Technology                |
| 104                  | Kavi                                     | IT                                     | Python                             | Information Technology                |
| 105                  | Rahul                                    | CSE                                    | Java                               | ComputerScience Engineering           |
| 106                  | Sneha                                    | ECE                                    | Python                             | Electronics Communication Engineering |
| 107                  | Meena                                    | ECE                                    | DBMS                               | Electronics Communication Engineering |
| 108                  | John                                     | IT                                     | Java                               | Information Technology                |
| 109                  | Akash                                    | CSE                                    | Python                             | ComputerScience Engineering           |
| 110                  | Divya                                    | ECE                                    | DBMS                               | Electronics Communication Engineering |
| 111                  | Harish                                   | IT                                     | Java                               | Information Technology                |
| 112                  | Pooja                                    | CSE                                    | Python                             | ComputerScience Engineering           |
| 113                  | Vicky                                    | ECE                                    | DBMS                               | Electronics Communication Engineering |
| 114                  | Nisha                                    | IT                                     | Java                               | Information Technology                |
| 115                  | Kiran                                    | CSE                                    | Python                             | ComputerScience Engineering           |
| 116                  | Sanjay                                   | ECE                                    | Java                               | Electronics Communication Engineering |
| 117                  | Anu                                      | IT                                     | DBMS                               | Information Technology                |

Example:

Department FullFormDepartment

|     |                                           |
|-----|-------------------------------------------|
| IT  | Information Technology                    |
| CSE | Computer Science Engineering              |
| ECE | Electronics and Communication Engineering |

Purpose of Dropdown Filter

The dropdown filter helps users:

- filter records easily
- display specific department data
- analyze variations dynamically

Example

When selecting:

IT

from the dropdown,

the output displays:

Information Technology

| Department           | Course               | FullFormDepartment     |
|----------------------|----------------------|------------------------|
| 1 distinct, 0 unique | 3 distinct, 1 unique |                        |
| IT                   | DBMS                 | Information Technology |
| IT                   | Python               | Information Technology |
| IT                   | Java                 | Information Technology |
| IT                   | Java                 | Information Technology |
| IT                   | Java                 | Information Technology |
| IT                   | DBMS                 | Information Technology |
| IT                   | DBMS                 | Information Technology |

Similarly,

| Selected Value | Output Displayed                          |
|----------------|-------------------------------------------|
| CSE            | Computer Science Engineering              |
| IT             | Information Technology                    |
| ECE            | Electronics and Communication Engineering |

| Department           | Course               | FullFormDepartment          |
|----------------------|----------------------|-----------------------------|
| 1 distinct, 0 unique | 3 distinct, 1 unique |                             |
| CSE                  | Python               | ComputerScience Engineering |
| CSE                  | Java                 | ComputerScience Engineering |
| CSE                  | Java                 | ComputerScience Engineering |
| CSE                  | Python               | ComputerScience Engineering |
| CSE                  | Python               | ComputerScience Engineering |
| CSE                  | Python               | ComputerScience Engineering |
| CSE                  | Python               | ComputerScience Engineering |

|                      |                      |                                       |
|----------------------|----------------------|---------------------------------------|
| 1 distinct, 0 unique | 3 distinct, 1 unique |                                       |
| ECE                  | Python               | Electronics Communication Engineering |
| ECE                  | DBMS                 | Electronics Communication Engineering |
| ECE                  | DBMS                 | Electronics Communication Engineering |
| ECE                  | DBMS                 | Electronics Communication Engineering |
| ECE                  | Java                 | Electronics Communication Engineering |
| ECE                  | Java                 | Electronics Communication Engineering |

## Conclusion

In this session, relationships between tables were successfully created in Power BI using Model View. Different relationship types such as One-to-One, One-to-Many, Many-to-One, and Many-to-Many were understood. Conditional Columns were also created to generate full department names automatically using conditions. These concepts are important for data modeling and data transformation in Power BI.